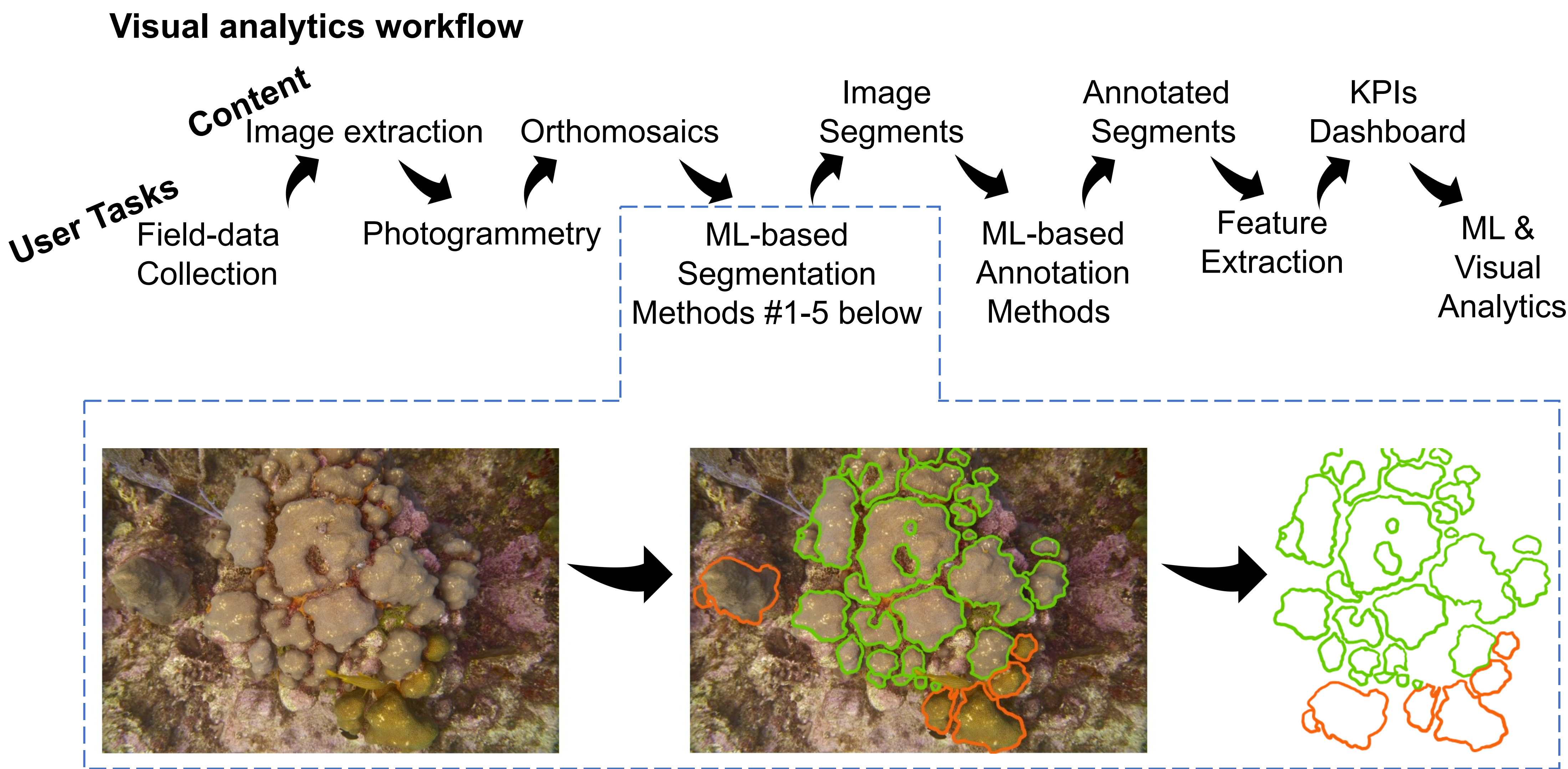
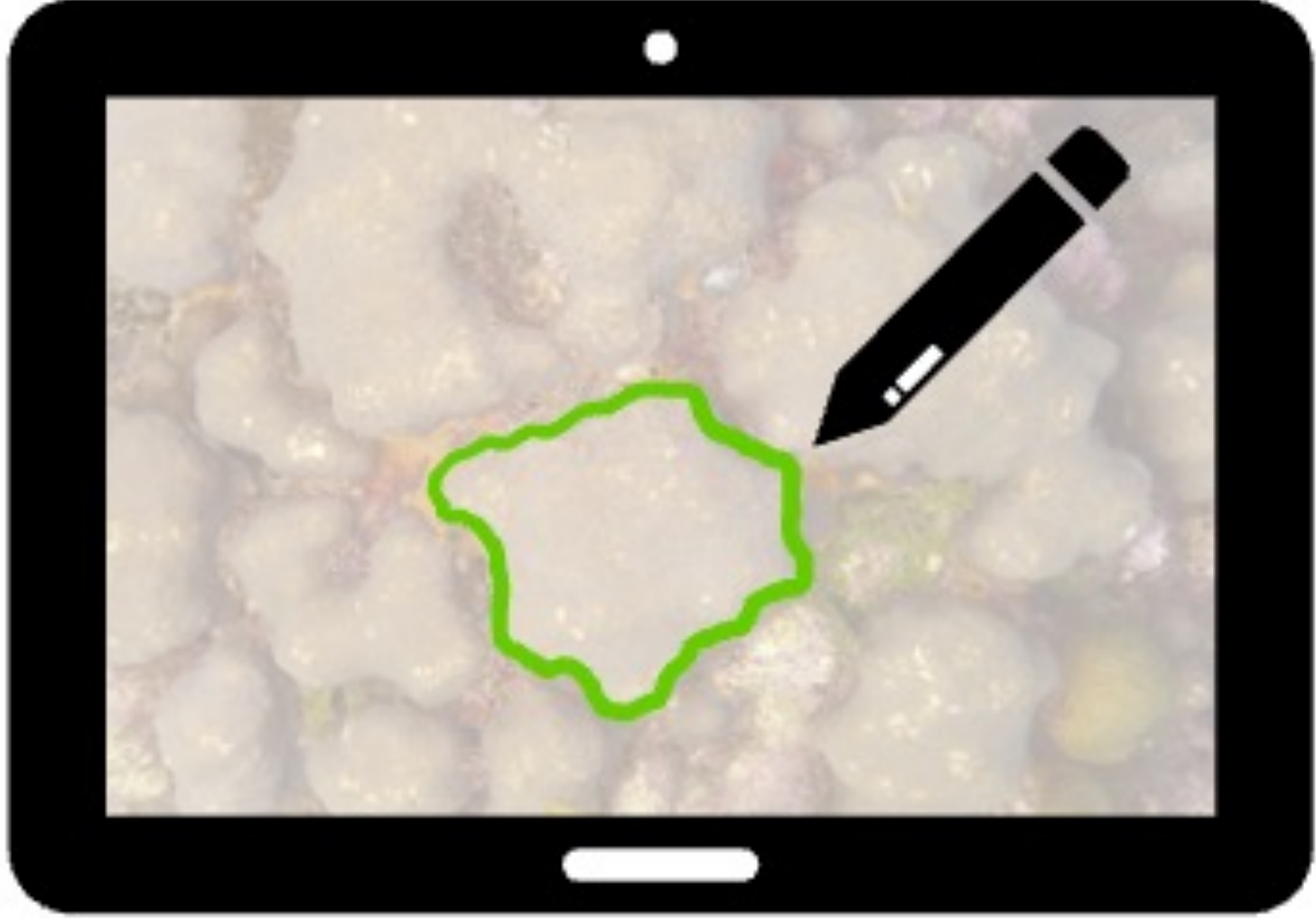
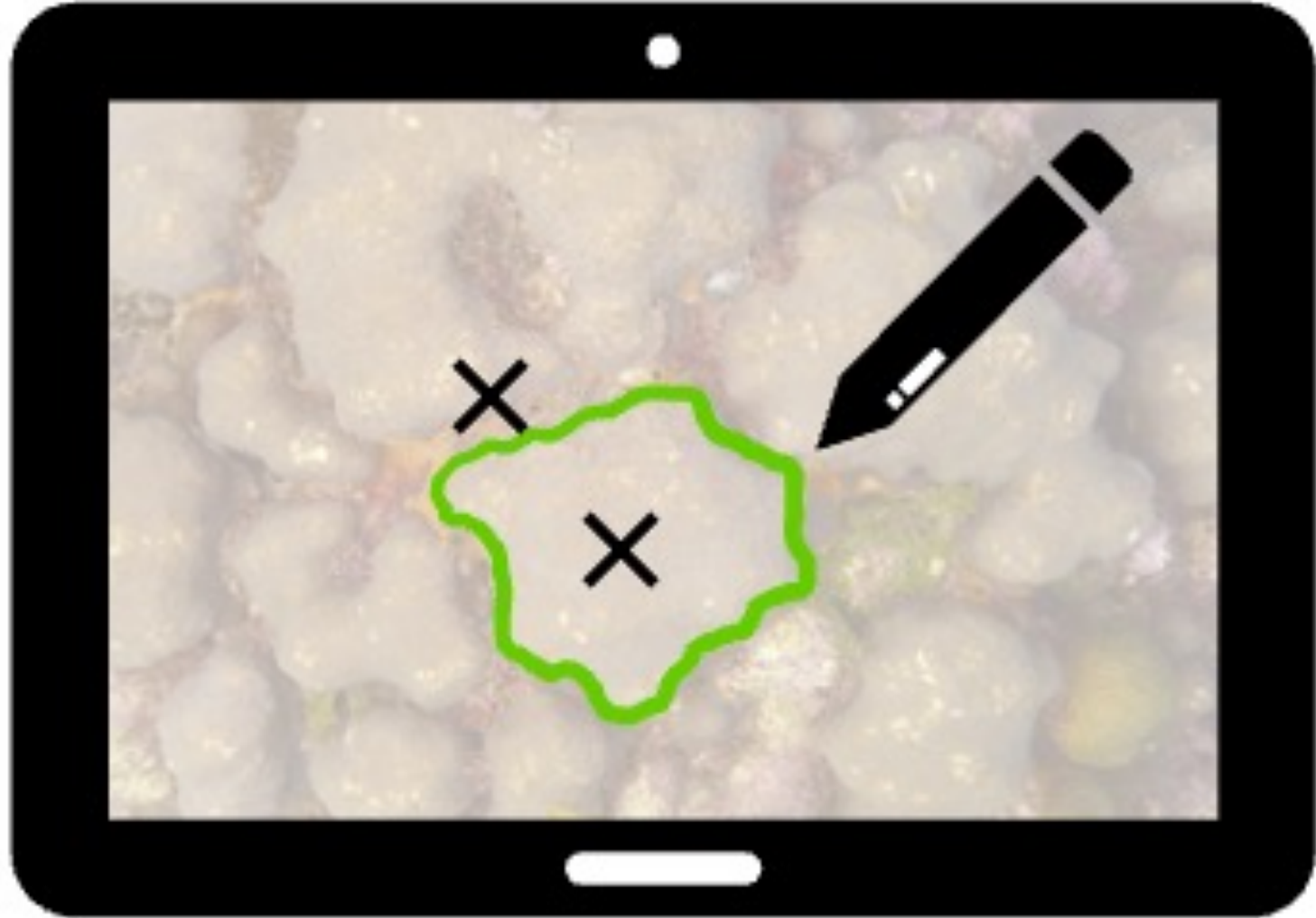
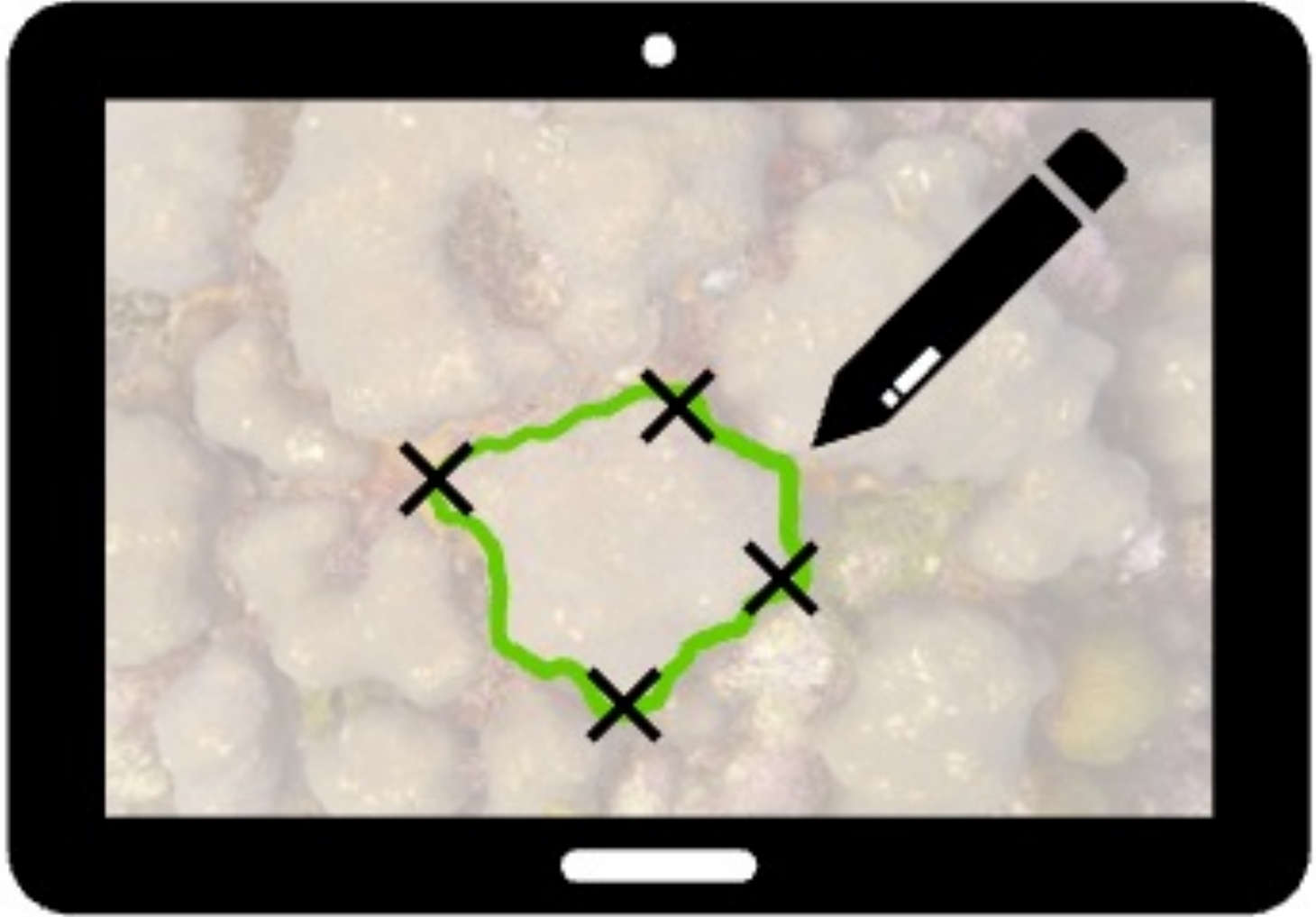
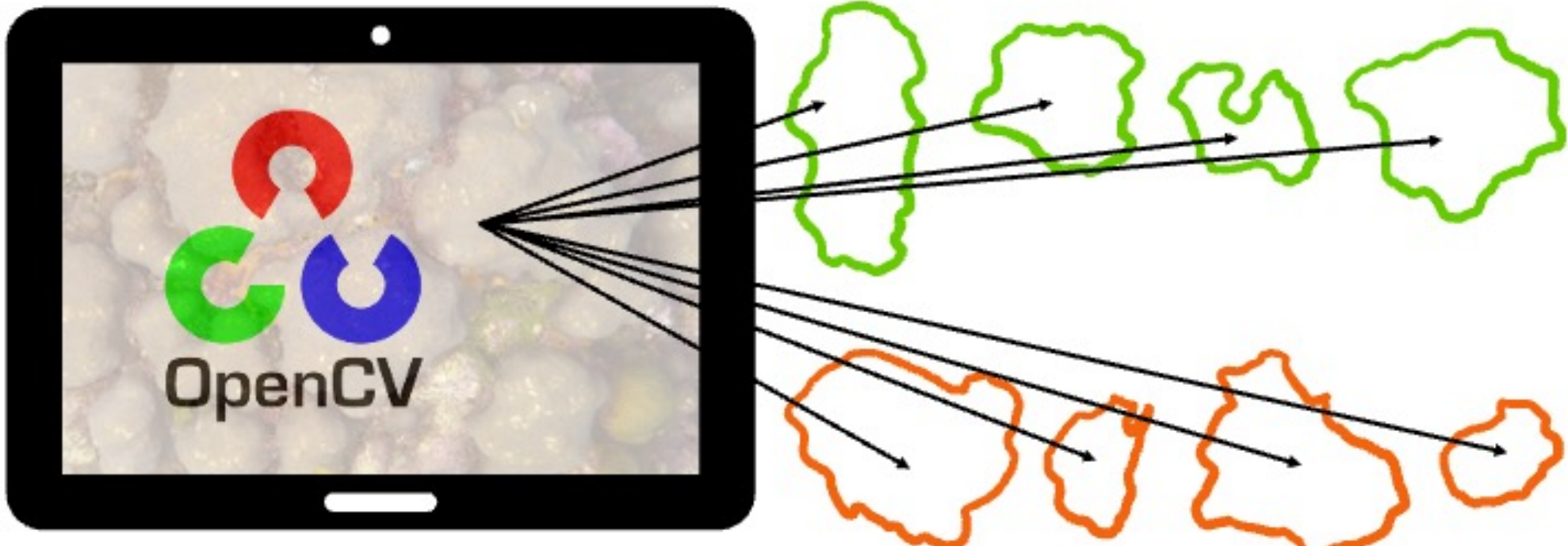


Abstract
AI and machine learning models are increasingly integrated into analytical workflows. These models can assist or replace many human-intensive tasks but also alter human-intensive interactions within redesigned workflows. In the context of orthomosaic image segmentation and classification, machine learning is shown to automate key tasks while requiring new interactions and selections from analysts. This work overviews the changing analyst views and interactions needed to leverage emerging machine learning models in orthomosaic analysis workflows.



Data • Orthomosaic of benthic zone • Billions of pixels, 1 km², mm to cm resolution KPIs and insights 1. Percent coral cover 2. Species biodiversity 3. Monitoring of outplanted corals 4. Monitoring of keystone species 5. Coral assemblage composition 6. Bleaching 7. Algal overgrowth	Method 1 • User interaction: manual segmentation • User interaction: manual annotation 	Method 2 • TagLab • User interaction: two clicks inside and outside • ML-based segmentation • User interaction: species classification from list 
Method 3 • TagLab • User interaction: four clicks in colony corners • ML-based segmentation • User interaction: species classification from list 	Method 4 • CV-based segmentation • User interaction: user selection of contours • User interaction: manual species classification 	Method 5 • ML-based segmentation via Segment Anything • Optional user interaction: filter the masks • ML-based classification via Convolutional Neural Network (CNN) or Vision Transformer (ViT) • User interaction: review of segments and labels 

Summary and Impacts

- Increasing ML-based automation, increasing scalability, and changing user tasks
- Nearing automation of image segmentation tasks
- Nearing automation of image classification tasks
- Reef analysis previously limited to km² scale due to human interaction
- Reef analysis now scalable to regional scales, new bottleneck limits of human-involved orthomosaic generation
- Changing scale of KPI and insights >> changing approaches to decision making, planning, mitigation, and adaptation

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